

CHAPTER 5

Associative Memories

Ex 5.1 BAM

Given - $x_1 = \begin{bmatrix} \quad \end{bmatrix}$ & $y_1 = \begin{bmatrix} \quad \end{bmatrix}$
 $x_2 = \begin{bmatrix} \quad \end{bmatrix}$ & $y_2 = \begin{bmatrix} \quad \end{bmatrix}$

Bipolar step function $g_h(z) = \begin{cases} +1, z \geq 0 \\ -1, z < 0 \end{cases}$

① Weight matrix

$$W = x_1^T y_1 + x_2^T y_2$$

② Correctness of recalls of the original pattern pairs.

$$\begin{matrix} g_h[x_1, W] \\ g_h[x_2, W] \end{matrix} \left\{ \begin{array}{l} \text{Forward} \\ \text{recalls} \end{array} \right\} \begin{matrix} g_h[y_1, W^T] \\ g_h[y_2, W^T] \end{matrix} \left\{ \begin{array}{l} \text{Backward} \\ \text{recalls} \end{array} \right\}$$

③ Correctness of recalls of the complementary pattern pairs

$$\begin{matrix} g_h[x_1^c, W] \\ g_h[x_2^c, W] \end{matrix} \left\{ \begin{array}{l} \text{Forward} \\ \text{recalls} \end{array} \right\} \begin{matrix} g_h[y_1^c, W^T] \\ g_h[y_2^c, W^T] \end{matrix} \left\{ \begin{array}{l} \text{Backward} \\ \text{recalls} \end{array} \right\}$$

④ Steady state recall pattern pair of the noisy vector, $x_t = \begin{bmatrix} \quad \end{bmatrix}$ & Given

$$\begin{matrix} g_h[x_t, W] = y_t \\ g_h[y_t, W^T] = x_1 \end{matrix} \rightarrow \begin{matrix} g_h[x_1, W] = y_1 \\ g_h[y_1, W^T] = x_1 \end{matrix}$$

$\therefore$ Steady state recall pattern pair is x_1 & y_1

Ex 5.2 TAM

Given:- Sequence of temporal patterns

$$x_1, x_2, x_3, x_1$$

$$x_1 = \begin{bmatrix} \quad \end{bmatrix} \quad x_2 = \begin{bmatrix} \quad \end{bmatrix} \quad x_3 = \begin{bmatrix} \quad \end{bmatrix}$$

① Memory matrix

$$W = x_1^T x_2 + x_2^T x_3 + x_3^T x_1$$

② Correctness of the original forward recalls and backward recalls.

Forward recalls

$$g_h[x_1, W] = x_2$$

$$g_h[x_2, W] = x_3$$

$$g_h[x_3, W] = x_1$$

Backward recalls

$$g_h[x_1, W^T] = x_3$$

$$g_h[x_3, W^T] = x_2$$

$$g_h[x_2, W^T] = x_1$$

③ Correctness of complementary forward recalls and complementary backward recalls.

Forward recalls

$$g_h[x_1^c, W] = x_2^c$$

$$g_h[x_2^c, W] = x_3^c$$

$$g_h[x_3^c, W] = x_1^c$$

Backward recalls

$$g_h[x_1^c, W^T] = x_3^c$$

$$g_h[x_3^c, W^T] = x_2^c$$

$$g_h[x_2^c, W^T] = x_1^c$$

④ Noisy pattern recalls, $x_t = \begin{bmatrix} \quad \end{bmatrix}$ & Given

Forward recall

$$g_h[x_t, W] = x_3$$

Backward recall

$$g_h[x_t, W^T] = x_1$$

$\therefore x_t$ is a corrupted version of x_2 .

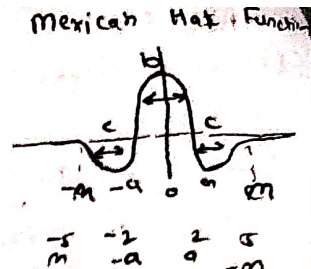
CHAPTER 6

Self-Organizing Maps

Ex 6.1

Given - $\alpha = 7$
 $S = 50$
 $m = 5$

$a = 2$
 $b = 0.06$
 $c = -0.04$



The lateral feedback weights of the Mexican hat function are given by

$$f_{-2j} = f_{-1j} = f_{0j} = f_{1j} = f_{2j} = 0.06 \text{ for } j=1 \text{ to } 11$$

$$f_{-5j} = f_{-4j} = f_{-3j} = f_{3j} = f_{4j} = f_{5j} = -0.04$$

	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}	y_{11}
$k-1$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$
k	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$	$\begin{bmatrix} \quad \end{bmatrix}$

Given

Computation of $y_j(k)$ for $j=1$ to 11 at any $k \geq 1$ is

$$y_j(k) = F \left[\alpha \sum_{-m}^m f_{mj} y_{j+m}(k-1) \right] - (A)$$

$$\begin{aligned} &= F \left[7 \left\{ f_{-5j} y_{j-5}(k-1) + f_{-4j} y_{j-4}(k-1) \right. \right. \\ &\quad + f_{-3j} y_{j-3}(k-1) + f_{-2j} y_{j-2}(k-1) + f_{-1j} y_{j-1}(k-1) \\ &\quad + f_{0j} y_{j+0}(k-1) + f_{1j} y_{j+1}(k-1) + f_{2j} y_{j+2}(k-1) \\ &\quad \left. \left. + f_{3j} y_{j+3}(k-1) + f_{4j} y_{j+4}(k-1) + f_{5j} y_{j+5}(k-1) \right\} \right] \end{aligned}$$

$$\begin{aligned} &= F \left[7 \left\{ -0.04 \times \underset{\substack{\uparrow \\ y_5}}{(0.0)} - 0.04 \times \underset{\substack{\uparrow \\ y_4}}{(0.0)} - 0.04 \times \underset{\substack{\uparrow \\ y_3}}{(0.0)} \right. \right. \\ &\quad + 0.06 \times \underset{\substack{\uparrow \\ y_2}}{(0.0)} + 0.06 \times \underset{\substack{\uparrow \\ y_1}}{(0.0)} + 0.06 \times \underset{\substack{\uparrow \\ y_0}}{(0.0)} \\ &\quad + 0.06 \times (y_1) + 0.06 \times (y_2) - 0.04 (y_3) \\ &\quad \left. \left. - 0.04 \times (y_4) - 0.04 \times (y_5) - 0.04 (y_6) \right\} \right] \\ &= F [-0.5318] \end{aligned}$$

$$y_j(k) = 0$$

$$* F[z] = \max(\min(z, S), 0) = \begin{cases} 0, & z < 0 \\ z, & 0 \leq z < S \\ S, & S \leq z \end{cases}$$

[Note] - $S=50$ in this example

Similarly solve for $j=2$ till $j=11$

To find $k+1$

$$y_j(k+1) = F \left[\alpha \sum_{-m}^m f_{mj} y_{j+m}(k) \right] - (B)$$

CHAPTER 7

Learning Vector Quantization

Ex 7.1

Given - Input			Initial weights		
k	x_k	T_k	j	w_j	C_j
1	[0 0]	1	1	[0 1]	1
2	[0 1]	1	2	[1 0]	2
3	[1 0]	2			
4	[1 0]	2			

Initial gain value $g(1) = 0.5$
 $g(t+1) = 0.5(g(t))$

Iteration 1: $g(1) = 0.5$

Presenting input $x_1 = [0 0]$ of class T_1

$$d_1 = (0-0)^2 + (0-1)^2 + (0-1)^2 = 1$$

$$d_2 = (0-1)^2 + (0-1)^2 + (1-0)^2 = 3$$

$\therefore d_1$ is smaller

$\therefore$ Weight vector w_1 of class C_1 will be updated

$\therefore$ Input x_1 and weight w_1 are of same class

$$w_j(t+1) = w_j(t) + g(t) [x_i - w_j(t)]$$

Input & weights are of same class

$$w_j(t+1) = w_j(t) + g(t) [x_i - w_j(t)]$$

Input & weights are of different class

$$w_1 = [0 \ 1 \ 1] + 0.5 \times ([0 0] - [0 1])$$

$$= [0 \ 0.5 \ 1]$$

Presenting input $x_2 = [0 1]$ of class T_1

with updated weights $w_1 = [0 \ 0.5 \ 1]$ &

$$w_2 = [1 \ 1 \ 0]$$

$$d_1 = (0-0)^2 + (1-0.5)^2 + (1-1)^2 = 0.25$$

$$d_2 = (0-1)^2 + (1-1)^2 + (1-0)^2 = 2$$

$\therefore d_1$ is smaller

$\therefore$ Weight vector w_1 of class C_1 will be updated

$\therefore$ Input x_2 and weight w_1 are of same class

$$w_1 = [0 \ 0.5 \ 1] + 0.5([0 1] - [0 \ 0.5 \ 1])$$

$$= [0 \ 0.75 \ 1]$$

Similarly solve for x_3 & x_4

After iteration 1

$$w_1 = [0 \ 0.75 \ 1]$$

$$w_2 = [1 \ 0.50 \ 0]$$

Then for iteration 2

$$\text{Compute } g(2) = 0.5 \times g(1) = 0.5 \times 0.5 = 0.25$$

CHAPTER 8

Radial Basis Function Networks

Ex 8.1

Given - Input			P	T_p
k	x_k	d_k		
1	[0, 0]	1	1	[0, 0]
2	[0, 0.9]	0	2	[1, 0]
3	[1, 0]	0		
4	[1, 0.9]	0		

$l_2 \text{ norm} = 1$

Using the Gaussian fn

$$y(x_k) = \sum_{p=1}^2 w_p \exp(-\|x_k - T_p\|^2) + w_0$$

To map input-output pair, we need

$$y(x_k) = d_k \text{ for } k=1,2,3,4$$

where x_k is input vector and d_k is desired output.

The radial basis function value at the p hidden unit due to k^{th} input pattern is

$$g_{kp} = \exp(-\|x_k - T_p\|^2) \text{ for } k=1,2,3,4$$

$$p=1,2$$

$$= \exp\{-[(x_{k1} - T_{p1})^2 + (x_{k2} - T_{p2})^2]\}$$

$$g_{21} = \exp\{-[(0-0)^2 + (0.9-0)^2]\}$$

$$= 0.444885$$

$$G = \begin{bmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \\ g_{31} & g_{32} \\ g_{41} & g_{42} \end{bmatrix}$$

Let the combined weight vector be

$$w_c = [w_0 \ w_1 \ w_2]^T$$

The following matrix eqⁿ can be obtained

$$w_c = H^+ D$$

where H^+ represents pseudo-inverse of matrix H

$$H = \begin{bmatrix} 1 & g_{11} & g_{12} \\ 1 & g_{21} & g_{22} \\ 1 & g_{31} & g_{32} \\ 1 & g_{41} & g_{42} \end{bmatrix}$$

$$2. H^+ = (H^T H)^{-1} H^T$$

$$w_c = H^+ D$$

$$w_c = [\quad]$$